

项目计划书 / PROJECT PLAN

AIFX Phase 2 Face Swap API - GCP 云端部署四天计划

中英混合版：面向 public API、API account、Cloud Run 部署与 ComfyUI 调用

准备对象 / Prepared for: Zooey / AIFX Studio Phase 2

准备日期 / Prepared on: 2026 年 7 月 6 日

执行周期 / Execution window: 4 working days

部署目标 / Deployment target: Google Cloud Platform: Cloud Run, Cloud Storage, Firestore, Secret Manager

不包含 / Excluded: 不包含 7 月 10-13 缓冲任务计划

执行摘要 / Executive Summary

本计划的目标是把 Phase 2 face swap API 完整部署到 GCP。系统将提供一个 public HTTPS endpoint, 但通过 API account / API key 控制访问。Cloud Run 负责运行 FastAPI 后端, Cloud Storage 保存 original/crop/ComfyUI output/final images, Firestore 保存 task history、API accounts、request logs 和 LoRA config。ComfyUI 已经准备好, 本服务只负责通过 API 调用它。

1. 目标结果 / Target Outcome

最终交付一个 GCP-hosted FastAPI service, 对外提供核心 Phase 2 endpoint:

Primary API / 核心接口

POST /api/v1/face-swap
Authorization: Bearer <API_ACCOUNT_KEY>

- Public HTTPS API: 外部测试方、ComfyUI 或 AIFX Studio 可以通过公网调用。
- API account authentication: 不是所有人都能访问, 必须携带有效 API key。
- Face detection and cropping: 复用 Phase 1 检测与裁剪逻辑, 后续可加入 MediaPipe + InsightFace fusion。
- LoRA mapping: 每张脸或每个 character_id 映射到对应 LoRA filename。
- ComfyUI orchestration: 动态注入 cropped face image 和 LoRA filename, 并调用外部 ComfyUI API。

- Spatial merging / blending: 把 ComfyUI 输出的人脸融合回原图，避免明显方框边缘。
- Cloud persistence: 所有图片和任务记录保存在 GCP 服务中。

2. GCP 架构 / Recommended Architecture

模块 / Layer	GCP Service	用途 / Purpose
API runtime	Cloud Run	运行 FastAPI face swap backend, 提供 public HTTPS API。
Image storage	Cloud Storage	保存 original images、crops、ComfyUI outputs、final merged images。
Metadata	Firestore	保存 task history、API accounts、request logs、LoRA config。
Secrets	Secret Manager	保存 ComfyUI endpoint/token、API account config 和服密。
Logs	Cloud Logging	查看 request failure、ComfyUI timeout、deployment issue。
Access control	IAM	授权 Cloud Run 访问 Storage、Firestore 和 Secrets。

边界说明 / Important Boundary

本计划部署的是 face swap API 和支撑数据层。ComfyUI 本体已准备好，作为 external service 调用，不纳入本次 GCP 部署范围。

3. API Account 设计 / Access Control Plan

API account 用于替代单一共享 token。每个调用方都可以拥有自己的 key、status、scope 和 request log。

Collection	主要字段 / Key Fields	说明 / Purpose
api_accounts	account_id, name, status, scopes, rate_limit_per_minute, created_at	表示外部用方，例如 Henry test account 或 ComfyUI worker account。
api_keys	account_id, key_prefix, key_hash, status, expires_at, last_used_at	只保存 hashed key；raw API key 只在创建时展示一次。
api_request_logs	account_id, endpoint, status_code, latency_ms, error_message, created_at	用于审计 public API 调用和排查错误。

- No Authorization header -> 401 Unauthorized.
- Invalid API key -> 401 Unauthorized.
- Disabled account -> 403 Forbidden.
- Missing face:swap scope -> 403 Forbidden.
- Valid account -> allowed to call /api/v1/face-swap and receive task_id + result URLs.

4. 四天计划 / Four-Day Implementation Plan

Day	目标 / Goal	主要工作 / Main Work	完成标准 / Definition of Done
Day 1 - Jul 6	GCP architecture + API contract	确定 /api/v1/face-swap request/response; 设计 API account collections; 创建 GCP resources: Cloud Run API, Storage bucket, Firestore, Secret Manager, service account。	GCP resources ready; API contract fixed; account model and deployment target documented。
Day 2 - Jul 7	API account + cloud persistence	实现 API account middleware; Cloud Storage adapter; Firestore task history/request logs; /api/v1/face-swap skeleton; 上传原图; 运行 Phase 1 detection。	Valid API key creates task; invalid access rejected; image/task records stored in GCP。
Day 3 - Jul 8	ComfyUI integration + merging	读取 ComfyUI workflow JSON; 动态注入 crop image 和 LoRA filename; 调用 ComfyUI; 多脸顺序处理; 保存 output; blend 回原。	Single-face E2E works; multi-face loop supported; final merged image stored and returned。
Day 4 - Jul 9	Cloud Run deploy + public API test	部署 FastAPI 到 Cloud Run; 配置 IAM/secrets; 测试 health、unauthorized、valid account、LoRA missing、ComfyUI timeout; 更新 README/handoff notes。	Cloud Run URL live; API account works; final_image_url returned; README includes deployment/testing steps。

5. 每日执行细节 / Daily Execution Details

Day 1 - Jul 6: GCP Foundation / 云端基础

- Enable Cloud Run, Cloud Storage, Firestore, Secret Manager, and Cloud Logging.
- Create Cloud Storage bucket for original/crop/ComfyUI output/final images.
- Create Firestore collections: task_history, api_accounts, api_keys, api_request_logs, lora_config.
- Define Cloud Run service account and IAM permissions.
- Freeze /api/v1/face-swap request/response contract.

Day 2 - Jul 7: Auth and Persistence / 认证与持久化

- Implement API account authentication with hashed API keys and scopes.
- Implement Cloud Storage upload paths grouped by account_id and task_id.
- Write Firestore task state transitions: pending, detecting, cropping, comfyui_processing, blending, completed, failed.
- Preserve original-image pixel coordinates from Phase 1 metadata.
- Return task_id, face list, bounding boxes, and stored image URLs.

Day 3 - Jul 8: ComfyUI and Blending / 工作流与融合

- Export ComfyUI workflow in API JSON format and identify dynamic input nodes.
- Inject cropped face image and LoRA filename into workflow payload.
- Submit to ComfyUI and monitor completion/timeout.
- Resize and blend each returned face into its original coordinates.
- Use PIL alpha feathering first; upgrade to OpenCV seamlessClone if needed.

Day 4 - Jul 9: Deployment and Review / 部署与测试

- Deploy service to Cloud Run using source deploy first; add Docker only if dependency installation fails.
- Set production environment variables and Secret Manager references.
- Run public curl tests with and without API account keys.
- Verify Cloud Storage URLs, Firestore task records, and Cloud Logging errors.
- Prepare handoff notes: API base URL, API account key, supported character IDs, known limitations.

6. 最终交付物 / Final Deliverables

交付物 / Deliverable	Required	说明 / Notes
Cloud Run FastAPI service	Yes	Public HTTPS API for Phase 2 face swap.
POST /api/v1/face-swap	Yes	Fully automated API entry point.
API account authentication	Yes	API keys with account status and scopes.
Cloud Storage integration	Yes	Original, crop, ComfyUI output, final image.
Firestore task history	Yes	Status tracking, logs, LoRA config.
ComfyUI API wrapper	Yes	Dynamic workflow JSON injection.
Blending module	Yes	No visible square boundaries around merged faces.
README and curl examples	Yes	Reviewer can call the public API.

7. 成功标准 / Success Criteria

- A public Cloud Run URL is available.
- Only valid API accounts can call /api/v1/face-swap.
- The API stores images and task records in GCP services.
- The API can call the existing ComfyUI service.
- The final merged image is returned as a URL.
- Failures are recorded without crashing the service.

Minimum Successful Phase 2 Version / 最小成功版本

一个部署在 GCP 的 public API，通过 API account 控制访问，可以接收图片、调用 ComfyUI、保存 artifacts，并返回 final merged image URL。